

AI Foundations | AIFo

Summary

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1. MACHINE LEARNING (ML)

There are many different AI research fields and AI applications. Algorithms and applications where a computer learns from data are typically in the field of statistical machine learning (ML).

Computer: performs a mapping from input to output.

Processing: any sort of non-trivial “calculation” (mapping) or information processing

Input: any sort of data/information (sensory input, bits, mechanical configuration...)

Output: any sort of response (data, actions, new state of a system...)

TODO:

W1 slides 2 14..17

2. PROBABILITY

TODO:

merge with MathFML, DigCod, WrStat

2.1. RANDOM VARIABLES

A random variable X is a function from the sample space to the real numbers.

$$X : S \rightarrow \mathbb{R}$$

Random Variables come in two flavours:

discrete: X takes any of a finite or countable set of values, e.g. $\{-8, 1.5, 1.5, 2.693, 5, 6.3, 10\}$

continuous: X takes any value of an uncountable range, e.g. the real numbers in the interval $(2, 7)$.

$\Pr(X = x)$ is the probability that the random variable X takes the value x . It's often written as $P(x)$ or $p(x)$ or $P_X(x)$.

TODO:

MathFML explained this a lot better

Definition 1: Range

The range of a random variable X , shown by $\text{Range}(X)$ or R_X , is the set of possible values of X .

2.2. PROBABILITY MASS FUNCTION (PMF)

The Probability Mass Function (PMF) of a discrete random variable is a function $P(x)$ that provides the probability for each value x of a discrete random variable X .

TODO:

CDF

TODO:

$$0 \leq P_X(x) \leq 1$$

$$\sum_{x \in R_X} P_X(x) = 1$$

$$\text{for any set } A \subset R_X, P(X \in A) = \sum_{x \in A} P_X(x)$$

Example 1:

Dice rolls

Value x of the random Variable X	1	2	3	4	5	6
$P(X = x)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

From the PMF, we can calculate the expected value

$$E[X] = \sum_{i=1}^{\infty} P(x_i) \cdot x_i$$

2.2.1. Multiple Random Variables

The (joint) properties of multiple random variables are defined by the **Joint Probability Mass Function**, usually simply called Joint Probability.

For **independent** random variables, the joint probability is simply the product of the individual probabilities

$$P(X \cap Y) = P(X) \cdot P(Y)$$

Example 2: Rolling two dice

The random experiment of rolling die 1 and die 2 together has 36 possible outcomes (events). The Joint Probability is:

$$P(X = 5, Y = 4) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

	$X = 1$	$X = 2$	$X = 3$	$X = 4$	$X = 5$	$X = 6$
$Y = 1$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 2$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 3$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 4$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 5$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 6$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$

2.3. CONDITIONAL PROBABILITY

For **dependent** random variables we use the conditional probability of X given Y defined as

$$P(X|Y) = \frac{P(X \cap Y)}{P(Y)}, \text{ when } P(Y) > 0$$

TODO:

$$P(X \cap Y) = P(X|Y) \cdot P(Y)$$

$$X \cap Y = \emptyset \Rightarrow P(X|Y) = 0$$

$$Y \subset X \Rightarrow P(X|Y) = 1$$

$$X \subset Y \Rightarrow P(X|Y) = \frac{P(X)}{P(Y)}$$

$$P(X \cap Y) = P(X)P(Y|X) = P(Y)P(X|Y)$$

Independence: Two events X and Y are independent iff $P(X \cap Y) = P(X)P(Y)$

Law of Total Probability: If $Y_1, Y_2, Y_3, \dots$ is a partition of the sample space S , then for any event X we have

$$P(X) = \sum_i P(X \cap Y_i) = \sum_i P(X|Y_i)P(Y_i)$$

Bayes' Rule: For any two Events X and Y , where $P(X) \neq 0$, we have

$$P(Y|X) = \frac{P(X|Y)}{P(X)}$$

Conditional independence: Two events X and Y are conditionally independent given an event Z with $P(Z) > 0$ if

$$\begin{aligned} P(X \cap Y|Z) &= P(X|Z)P(Y|Z) \\ \Rightarrow P(X|Y \cap Z) &= P(X|Z) \end{aligned}$$

Example 3:

TODO:

2.4. PROBABILITY DENSITY FUNCTION (PDF)

$$P(-\infty < x < \infty) = 1$$

$$P(a < x < b) = \int_a^b f(x) dx$$

$$P(x) = 0$$

TODO:

https://en.wikipedia.org/wiki/68%E2%80%9393.95%E2%80%9399.7_rule

TODO:

MathFML

2.5. DISTRIBUTIONS

2.5.1. Bernoulli distribution

A random variable X is said to be a Bernoulli random variable with parameter p , shown as $X \sim \text{Bernoulli}(p)$, if its PMF is given by

$$P_X(x) = \begin{cases} p & \text{for } x = 1 \\ 1 - p & \text{for } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

where $0 < p < 1$

A Bernoulli random variable is associated with a certain event A . If event A occurs, then $X = 1$; otherwise $X = 0$. For this reason the Bernoulli random variable is also called the indicator random variable. The indicator random variable I_A for an event A is defined by

$$I_A = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

The indicator random variable for an event A has Bernoulli distribution with parameter $p = P(A)$, so we can write $I_A \sim \text{Bernoulli}(P(A))$.

TODO:

https://www.probabilitycourse.com/chapter3/3_1_5_special_discrete_distr.php